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QSAR study on melanocortin-4 receptors by support vector machine

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ABSTRACT

Quantitative structure activity relationship (QSAR) of the melanocortin-4 receptor (MC4R) binding affinities (K_i) of trans-4-(4-chlorophenyl) pyrrolidine-3-carboxamides of piperazinecyclohexanes was studied. A suitable set of molecular descriptors was calculated and the genetic algorithm (GA) was employed to select those descriptors that resulted in the best-fit models. The multiple linear regression (MLR), and the support vector machine (SVM) were utilized to construct the linear and nonlinear QSAR models. The models were validated using Leave-One-Out (LOO) and Leave-Group-Out (LGO) cross-validation, external test set, and chance correlation. The SVM model generalizes better than the MLR model. The SVM model, with high statistical significance ($R^2_{\text{train}} = 0.908$, $Q^2_{\text{LOO}} = 0.781$, $Q^2_{\text{LGO}} = 0.872$), could be used to predict melanocortin-4 receptor binding affinities of piperazinecyclohexanes.

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1. Introduction

The melanocortin-4 receptor (MC4R) is a member of the G-protein-coupled receptor super family, and plays an important role in biological functions including regulation of feeding behavior [1]. MC4R agonists have been extensively studied in an effort to discover small molecules for the treatment of obesity [2], and several small molecule MC4R agonists from different chemical classes have been reported [3,4]. MC4R antagonists, on the other hand, have been shown to reverse lean body mass loss and increase food intake in animal models of cachexia, suggesting potential in cancer cachexia treatment [5,6]. Originally, Pyrrolidine derivatives were reported by Ujjainwalla as potent and selective MC4R agonists [7]. Recently, Caroline W. Chen et al. synthesized the trans-4-(4-chlorophenyl) pyrrolidine-3-carboxamides of piperazinecyclohexanes as ligands for the MC4R [8]. The synthesized compounds were tested in competition binding assays, using HEK293 cells expressing the human MC4R and [¹²⁵I]-NDP-MSH as the radio labeled ligand [9].

Experimentally determined binding affinity (K_i) values are not always available from literature sources, and estimated values are

often employed instead. Therefore, it is of interest to develop methods for estimating the MC4R binding affinity of compounds.

Quantitative structure activity relationship (QSAR) method provides a promising approach for the estimation of the affinity based on the descriptors solely derived from the molecular structures. The QSAR method, has found diverse applications for predicting compounds' properties, including biological activity prediction [10,11], physical property prediction [12,13] and binding affinity [14,15]. In QSAR studies, there are some linear and nonlinear techniques which can be applied for the construction of model, such as multiple linear regression (MLR), partial least-squares (PLS), artificial neural networks (ANN) and support vector machine (SVM). Although the collinearity problem of the MLR method can be overcome through the development of PLS method and this method has shown to be an efficient approach in the monitoring of many complex processes, reducing the high dimensional strongly cross-correlated data to a much smaller and interpretable set of principal components. But the MLR method yields simpler and easier to interpret linear models than those resulted by PLS. Also, the flexibility of neural networks enables them to discover more complex nonlinear relationships in experimental data [16]. Neural networks have some problems inherent to its architecture such as; overtraining, overfitting, network optimization, and reproducibility of results, due to random initialization of the networks and variation of stopping criterias [17]. Owing to these reasons, there is a tendency to use more accurate and

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informative techniques in QSAR analysis. The SVM is a new algorithm developed from the machine learning community [18]. SVM approach automatically controls the flexibility of the resulting classifier on the training data. Accordingly, by the design of the algorithm, the deteriorating effect of the input dimensionality on the generalization ability is largely suppressed. Due to its remarkable generalization performance, the SVM has attracted attention and gained extensive application, such as pattern recognition problems [17] drug design [19] QSAR [20] and quantitative structure–property relationship (QSPR) analysis [21]. In most of these cases, performance of SVM modeling either matches or is significantly better than that of traditional machine learning approaches.

The application of these techniques usually requires variable selection for building well-fitted models. There have been many variable selection methods the mostly used ones are stepwise regression, evolutionary algorithms and genetic algorithms and so on. A limitation of the stepwise regression search approach is that it presumes that there is a single “best” subset of X variables and seeks to identify it. There is often no unique “best” subset. All the possible regression models with a similar number of X variables to the stepwise regression solution should be subsequently fitted to study whether some other X variables subsets might be better. Nowadays, GA is well-known as an interesting and more widely used variable selection method [22–24]. GA is a stochastic method to solve the optimization problems defined by fitness criteria, applying the evolution hypothesis of Darwin and different genetic functions, i.e. crossover and mutation.

The main aim of the present work is to establish a new QSAR model for predicting binding affinity of the trans-4-(4-chlorophenyl) pyrrolidine-3-carboxamides of piperazinecyclohexanes as ligands for the melanocortin-4 receptor using the GA-SVM technique. Performance of this model was compared with that of the GA-MLR method.

2. Materials and methods

2.1. Data set

The data used in this QSAR study consisted of MC4R binding affinities (K_i), of trans-4-(4-chlorophenyl) pyrrolidine-3-carboxamides of piperazinecyclohexanes that have been reported by Caroline W. Chen et al. [8]. The activity data [K_i (nM)] for 4-alkylamino-6-(2-hydroxyethyl)-2-methylthiopyrimidines (Table 1) was converted to the logarithmic scale [pK_i (M)] and then used for subsequent QSAR analyses as the response variables.

2.2. Descriptor calculations and selection

The main step in every QSAR study is choosing and calculating the structural descriptors as numerical encoded parameters representing the chemical structures. In the present work the chemical structure of the molecules was constructed using Hyperchem package. AM1 semi-empirical calculation was used to optimize the 3D geometry of the molecules [25,26]. To ensure obtaining structure with optimum geometry, optimization was repeated many times with different starting geometries. The optimization was preceded by the Polak–Rebiere algorithm to reach 0.01 root mean square gradient. The Z-matrices of the structures were constructed by the software and then transferred to the Gaussian 98 program [27]. HyperChem, Gaussian 98 and Dragon packages [28] were used for calculation of molecular descriptors. Highest occupied molecular orbital (HOMO) and lowest unoccupied molecular orbital (LUMO) energies and molecular dipole moment were calculated by Gaussian 98. Quantum chemical indices of hardness (η); softness (S); electronegativity (χ); and electrophilicity (ω) were calculated according to

the equations proposed by Thanikaivelan et al. [29]. Some chemical parameters including molar volume (V), molecular surface area (SA), hydrophobicity ($\log P$), hydration energy (HE) and molecular polarizability were calculated using Hyperchem software. DRAGON software was used to calculate the descriptors, belonging to 18 different types of theoretical descriptors for each molecule. Descriptors with constant or almost constant values for all molecules were eliminated. Also, pairs of variables with a correlation coefficient greater than 0.90 were classified as intercorrelated, and only one of them was considered in developing the model. A total of 506 descriptors were considered for further investigations after discarding the descriptors with constant and intercorrelated ones.

MLR was utilized as linear technique, whereas SVM was employed as a nonlinear feature mapping technique for the construction of the QSAR models in this work. Since the SVM method cannot select the most significant descriptors from the pool of the calculated molecular descriptors, it would be necessary to use a variable selection method. In the present work, the genetic algorithm (GA) variable subset selection method was used for the selection of the most relevant descriptors from the pool of the remaining 506 descriptors. These descriptors would be used as inputs of the MLR and SVM.

3. Results and discussion

For the selection of the most important descriptors, genetic algorithm variable subset selection method based MLR was used. According to rule of thumb [30,31] at least five data points (compounds) should be included in the equation for every parameter (descriptor). On the other hand, the ratio of 5 training molecules for each descriptor must be included in the equation. In order to investigate the optimum number of descriptors to be used in a model for modeling pK_i , we have plotted a graph between numbers of descriptors against statistical parameters (R^2 , Q^2 and s). Fig. 1 shows the plots of R^2 , Q^2 and s as a function of the number of descriptors for the 1–12 parameter models. R^2 and Q^2 increased by increasing the number of descriptors. However, the values of s decreased with the increasing the number of descriptors. A perusal, Fig. 1 indicates that R^2 , Q^2 and s remain almost parallel to the X-axis (number of descriptors) after eight parameters and higher order models. This shows that the most suitable models are eight parametric models. We have carried out multiple regression analysis on that data set of 73 compounds and 8 descriptors. Since co-linearity between the variables degrades the performance of the MLR-based QSAR models, before a multiparametric analysis was undertaken, the correlation between each one of the variables used in this study and the pK_i data were examined. The correlation matrix itself exhibits how the used descriptors were mutually correlated and how they correlate with pK_i . The correlation matrix obtained in the present case is shown in Table 2. From Table 2, it could be seen that the correlation coefficient value of each pair of descriptors was less than 0.52, which meant that the selected descriptors were independent.

The selected variables are MSD, BELe5, JGI9, RDF040m, RDF010e, C-008, C-026, and N-075. As can be seen, there is not any selection for quantum chemical descriptors.

To examine the relative importance as well as the contribution of each descriptor in the model, for each descriptor the value of the mean effect (MF) was calculated. This calculation was performed with the equation below:

$$MF_j = \frac{\beta_j \sum_{i=1}^n d_{ij}}{\sum_{j=1}^m \beta_j \sum_{i=1}^n d_{ij}} \quad (1)$$

MF_j represents the mean effect for the considered descriptor j , β_j is the coefficient of the descriptor j , d_{ij} stands for the value of the

target descriptors for each molecule and, eventually, m is the descriptors number in the model. The MF value indicates the relative importance of a descriptor compared with the other descriptors in the model. Its sign exhibits the variation direction in

the values of the activities as a result of the increase (or reduction) of this descriptor values. The mean effect values are 0.454, 0.850, –0.144, –0.090, –0.106, 0.016, 0.028, and –0.007 for MSD, BELe5, JGI9, RDF040m, RDF010e, C-008, C-026, and N-075 respectively.

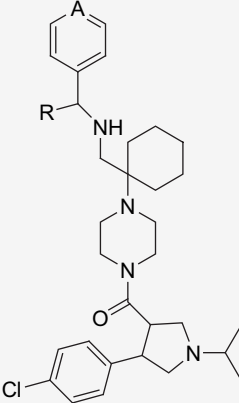
Table 1

Experimental and predicted values of pK_i for various piperazinecyclohexanes for training and test sets by GA-MLR and GA-SVM models.^a

Compound	Structure	R	A	pK_i	Pred.(MLR)	Pred.(SVM)
1		H	–	5.92	6.75	6.01
2		PhCH ₂ CO	–	6.29	6.36	6.38
3 ^a		PhCH ₂ SO ₂	–	6.47	6.49	6.87
4		PhSO ₂	–	6.48	6.6	6.41
5		MeOCH ₂ CH(Me)	–	6.51	6.34	6.46
6		CyclohexaneCH ₂	–	6.52	7.03	6.61
7		PhCH ₂	–	7.59	7.25	7.50
8		2-FuranCH ₂	–	7.42	7.04	7.33
9 ^a		3-FuranCH ₂	–	7.12	7.81	7.62
10		2-ThiopheneCH ₂	–	7.43	7.47	7.34
11		3-ThiopheneCH ₂	–	7.66	7.55	7.57
12		2-ThiazoleCH ₂	–	6.89	6.35	6.80
13		4-ImidazoleCH ₂	–	6.66	6.76	6.75
14		2-ImidazoleCH ₂	–	6.25	6.37	6.34
15		1-Me-2-imidazoleCH ₂	–	6.41	6.39	6.50
16		1,5-(Me)2-4-pyrazoleCH ₂	–	7.96	7.08	7.64
17		5-PyrimidineCH ₂	–	6.66	7.14	6.75
18		2-PyCH ₂	–	7.12	6.97	7.21
19		3-PyCH ₂	–	7.46	7.08	7.32
20		4-PyCH ₂	–	7.57	6.79	7.12
21		1-Me-2-indoleCH ₂	–	5.92	6.08	7.03
22		2-PyCH ₂ CH ₂	–	6.16	6.4	6.25
23		PhCH ₂	CH	6.85	6.86	6.76
24		CH(Me)CH ₂ OMe	CH	7	7.3	7.38
25		Cyclopentyl	CH	7.77	6.88	7.33
26		Cyclohexyl	CH	8.26	7.95	7.70
27		4-Tetrahydropyran	CH	7.54	7.98	7.63
28		MeCO	CH	6.72	6.34	6.63
29 ^a		EtCO	CH	6.68	6.52	6.73
30		nPrCO	CH	6.72	7.29	6.81
31		i-PrCO	CH	6.68	7.11	6.77
32 ^a		CyclopentaneCO	CH	6.74	6.92	6.82
33		MeNHCH ₂ CO	CH	7	6.23	6.91
34		MeCH(NH ₂)CO	CH	6.37	6.98	6.46
35		NH ₂ CH ₂ CH ₂ CO	CH	6.96	6.53	6.87
36 ^a		i-PrCH(NH ₂)CO	CH	6.17	6.88	6.52
37		PhCH ₂	N	6.7	6.5	6.61
38		H	N	5.66	6.05	5.71
39 ^a		MeCO	N	5.52	5.37	5.63
40		EtCO	N	5.51	5.61	5.60
41		EtNHCO	N	5.34	5.04	5.43
42		MeSO ₂	N	5.47	6.09	5.56
43 ^a		–	2-F	7.92	8.06	7.47
44		–	2-Cl	7.6	7.34	7.46
45		–	2-MeO	7.21	7.84	7.30
46		–	2-CF ₃	6.85	6.6	6.76
47 ^a		–	2-CF ₃ O	6.96	7.56	7.24
48 ^a		–	3-F	7.62	8.02	7.46
49 ^a		–	3-MeO	7.52	7.74	7.38
50 ^a		–	4-F	7.36	8.01	7.48
51		–	4-Cl	7.05	7.27	7.14
52 ^a		–	4-MeO	8.44	7.95	7.52
53		–	4-CF ₃	6.89	7.1	6.98
54		–	4-NO ₂	7.28	7.8	7.37
55		–	4-AcNH	7.54	7.81	7.45
56		–	4-Me ₂ N	7.19	7.52	7.28
57		–	4-COOMe	7.49	7.28	7.40
58		–	4-MeS	7.57	7.29	7.48
59		–	4-i-PrO	8.68	8.31	8.03
60		–	2,4-F	7.96	8	7.87
61		–	2,4-MeO	7.8	7.92	7.71
62 ^a		–	2-F,4-MeO	8.74	8.32	7.96
63		–	3,4-OCH ₂ O	8.08	7.4	7.79
64		–	3-F,4-MeO	9.3	8.01	8.11

(continued on next page)

Table 1 (continued)

Compound	Structure	R	A	pK _i	Pred.(MLR)	Pred.(SVM)
65		Me	COMe	6.96	7.41	7.05
66		Et	COMe	6.72	6.79	6.68
67		i-Bu	COMe	5.74	5.94	5.83
68		4-MeOC ₆ H ₄ CH ₂	COMe	5.55	5.47	5.52
69		MeCO	COMe	5.89	6.26	5.98
70		Me	N	6.12	6.28	6.21
71 ^a		Et	N	5.89	6.83	6.24
72		i-Bu	N	5.57	5.55	5.66
73 ^a		MeCO	N	5.82	5.84	6.04

^a The test set.

Besides demonstrating statistical significance, QSAR models should also provide useful chemical insights for drug design. For this reason, an acceptable interpretation of the QSAR results is provided below. By interpreting the descriptors contained in the model, it is possible to gain some insights into factors that relate to the binding affinity.

MSD is one of the topological descriptors, which has appeared in the model. MSD is the mean square distance index also called Balaban. The MSD is calculated from the second order distance distribution moment. The distance distribution moments are the moments for the distribution of topological distances in a molecular graph. The MSD index decreases with increasing the molecular branching in an isomeric series [32]. The mean effect value for this descriptor is 0.454. The high value of mean effect for MSD shows the significant effect of this descriptor in model. MSD displays a positive sign, which indicates that the pK_i is directly related to this descriptor. From above, it was concluded that by increasing the molecular branch, the pK_i value decreases.

BELe5 (lowest eigenvalue n. 5 of Burden matrix/weighted by atomic Sanderson electronegativities) is the second descriptor, appearing in the model. It is one of the BCUT descriptors. The BCUT (Burden, CAS, University of Texas) descriptors are the eigenvalues of a modified connectivity matrix known as the Burden matrix [33].

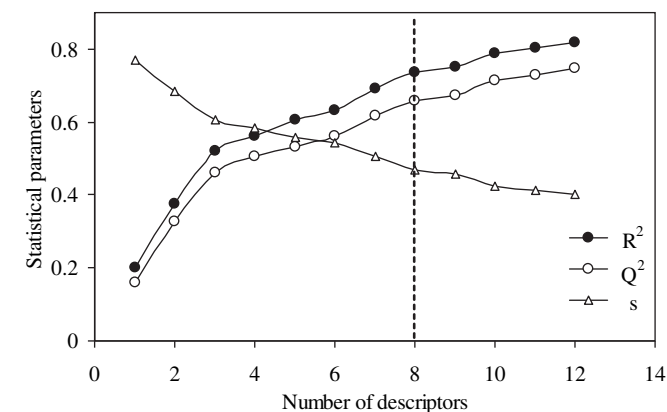


Fig. 1. Influences of the number of descriptors on: R², Q² and s, of the regression model.

The BELe5 has a highest mean effect value with positive sign. It can be concluded that BELe5 displays a great effect in the model and the electro-negativity of molecules is directly related to pK_i values.

The third descriptor is JGI9 (mean topological charge index of order 9). Topological charge indices were proposed to evaluate the charge transfer between pairs of atoms, and therefore the global charge transfer in the molecule [32]. The JGI9 is closely related to the molecular branching.

The fourth and fifth descriptors are RDF040m and RDF010e. They belong to the radial distribution function (RDF) descriptors. RDF in these forms meets all the requirements for the 3D structure descriptors. It is independent of the atom number (i.e. the size of a molecule), it is unique regarding the three-dimensional arrangement of the atoms and it is invariant against the translation and rotation of the entire molecule. Additionally, the RDF descriptors can be restricted to specific atom types or distance ranges to represent specific information in a certain three-dimensional structure space (e.g. to describe the steric hindrance or the structure/activity properties of a molecule). The radial distribution function (RDF) descriptors are based on the distance distribution in the molecule. The radial distribution function of an ensemble of *n* atoms can be interpreted as the probability distribution of finding an atom in a spherical volume of radius, *R*. In these descriptors (RDF040m and RDF010e) weighting schemes are the atomic masses and the atomic Sanderson electronegativities respectively, that show the mass and the electro-negativity of the molecules play a main role in these descriptors. RDF035m and RDF030v display negative signs, which indicates that the pK_i is inversely related to these descriptors. From above, it was concluded that by increasing the molecular mass and molecular electro-negativity the value of these descriptors increased, causing a reduction in their pK_i value.

The other descriptors (C-008, C-026, N-075) belong to the atom-centered fragment descriptors that describe each atom by its own atom type and the bond types and atom types of its first neighbors. Functionalities in a molecule can be represented by two to five atoms (corresponding to one bond to 4 bonds), which consist of a central atom and its neighboring bonded atoms. Hydrogen atoms are not considered [32].

The C-008, C-026 and N-075 descriptors, represent CHR₂X, R-CX-R and R-N-R/R-N-X respectively. C-008 and C-026 display a positive sign whilst N-075 displays a negative sign, thus

Table 2

Correlation coefficient matrix of the selected descriptors.

	MSD	BELe5	JGI9	RDF040m	RDF010e	C-008	C-026	N-075
MSD	1	−0.510	−0.260	−0.516	−0.277	0.053	−0.029	−0.098
BELe5		1	0.434	0.371	0.425	0.123	0.240	−0.040
JGI9			1	0.143	0.260	0.216	0.491	−0.249
RDF040m				1	0.140	0.208	0.202	−0.097
RDF010e					1	0.144	0.114	−0.385
C-008						1	0.301	−0.294
C-026							1	−0.325
N-075								1

indicating that the pK_i is directly related to the C-008 and C-026 descriptors while it is inversely related to the N-075 descriptor. It was concluded that by increasing the number of CHR₂X and R-CX-R substations of molecules the pK_i value increases, and by increasing the number of R-N-R or R-N-X substitution of molecules the pK_i value decreases.

In summary it is concluded that; branching, electro-negativity, mass, and the CHR₂X, R-CX-R and R-N-R substations of molecules play a main role in the binding affinities of the studied compounds. Also the Caroline W. Chen et al. [8] finding confirm some of our results. For example Caroline et al. found that the secondary amines or basic amines are important for affinity binding and we found that R-N-R substations of molecules is important for affinity binding. The N-group on pyrrolidine was studied by Caroline W. Chen et al. and they suggested that the basic nitrogen and a small lipophilic group such as isopropyl are important for binding affinity of these compounds. This result confirms our finding concerning with CHR₂X and R-CX-R substations.

3.1. Results of the MLR method

In order to build and test the model, a data set of 73 compounds was randomly separated into a training set of 58 compounds, which was used to build the model and a test set of 15 compounds, which was applied to test it. With the selected eight descriptors, we have built the linear model using the training set data, and the following equation was obtained:

$$pK_i = -16.251(\pm 4.498) + 47.465(\pm 12.437)MSD + 12.540(\pm 1.759)BELe5 - 286.790(\pm 64.271)JGI9 - 0.125(\pm 0.027)RDF040m - 0.213(\pm 0.089)RDF010e + 0.439(\pm 0.138)C-008 + 0.462(\pm 0.110)C-026 - 0.908(\pm 0.170)N-075 \quad (2)$$

The built model was used to predict the test set data. The prediction results are given in Table 1 and shown in Fig. 2. The square correlation coefficient R^2 was obtained to be 0.725 for the training set and 0.795 for the test set with root mean square error (RMSE) of 0.435 and 0.474, respectively.

The models obtained were validated using leave-one-out (LOO) and leave-group-out (LGO) cross-validation process. For LOO cross-validation, a data point is removed from the set, and the model is recalculated. The predicted activity for that point is then compared to its actual value. This is repeated until each data point is omitted once. For LGO, a 20% of the data points are removed from the data set and the model refitted; the predicted values for those points are then compared to its experimental values. Again, this is repeated until each data point has been omitted once. Cross-validation parameters are shown in Table 3. This indicates that the obtained regression model has a good internal and external predictive power.

The robustness of the resultant model was also validated with the chance correlation procedure. For a set of the trans-4-(4-chlorophenyl) pyrrolidine-3-carboxamides of piperazinecyclohexanes, the pK_i values were randomly attributed to the molecules. Then, the MLR modeling with the selected descriptors were repeated with the randomized data [34]. The randomization was repeated 20 times. The squared correlation coefficient of the model with the randomly selected data (R^2_{CR}) was used as statistical criterion (Table 3). If the R^2_{CR} values of these models were much lower than those of the original model, it could be considered that the model was reasonable, and had not been obtained by the chance.

3.2. Results of the SVM method

In order to develop a more accurate model, SVM was used to develop a model by the training set compounds as same as our previous works [35–37]. A detailed description of SVM theory can be found in several excellent books and tutorials [38–41]. LOO cross-validation method implied in SVM was used to build the model. Performance of SVM for regression depends on the combination of various factors. They are: kernel function type, capacity parameter C, ϵ of ϵ -insensitive loss function, and its corresponding parameters.

Firstly, the kernel function should be decided, which determines the sample distribution in the mapping space. The radial basis function (RBF) is commonly used in many studies because of its good general performance, and few parameters need to be adjusted [42]. In this work, the radial basis function was used, the form of which in R is as follows:

$$\exp(-\gamma^*|u-v|^2)$$

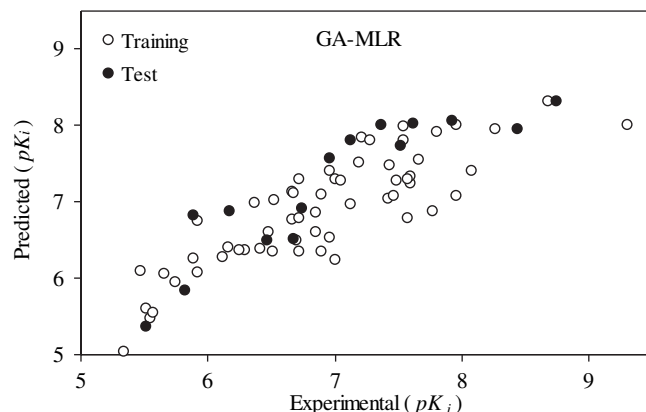


Fig. 2. Observed vs. calculated pK_i for the GA-MLR model with selected descriptors.

Table 3
Statistical results of GA-MLR and GA-SVM models.

	Training set			Test set		Whole data set					
	RMSE	R^2	Q^2_{LOO}	RMSE	R^2	RMSE	R^2	$F_{8, 64}^a$	P -value	Q^2_{LGO}	R^2_{CR}
GA-MLR	0.435	0.725	0.658	0.474	0.795	0.444	0.731	22.185	$<10^{-5}$	0.711	0.183
GA-SVM	0.279	0.908	0.781	0.410	0.853	0.310	0.892	37.522	$<10^{-5}$	0.872	0.215

^a F variance ratio of calculated and observed values with two degrees of freedom.

Where γ is a parameter of the kernel; u and v are two independent variables.

Secondly, corresponding parameters, i.e. γ of the kernel function greatly affect the number of support vectors, which has a close relation with the performance of the SVM and training time. Too many support vectors could produce overfitting and increase the training time. In addition, γ controls the amplitude of the RBF function and, therefore, controls the generalization ability of SVM.

Parameter ε -insensitive prevents the entire training set meeting boundary conditions and so allows for the possibility of sparsity in the dual formulation's solution. The optimal value for ε depends on the type of noise present in the data, which is usually unknown.

Lastly, the effect of capacity parameter C was tested. It controls the trade-off between maximizing the margin and minimizing the training error. If C is too small then insufficient stress will be placed on fitting the training data. If C is too large then the algorithm will overfit the training data. However, Ref. [42] indicated that the prediction error was scarcely influenced by C . To make the learning process stable, a large value should be set up for C initially.

We performed leave-one-out cross-validation to select the optimum values of the parameters. Finally the optimum values of γ , ε and C were fixed to 2, 0.09 and 6, respectively. It gave RMSE of 0.279 binding affinity units for the training set, 0.410 for the test set, and the corresponding correlation coefficients (R^2) were 0.908 and 0.853, respectively. For the optimal model, the Q^2_{LOO} and Q^2_{LGO} cross-validated coefficients were 0.781 and 0.822.

The calculated pK_i values obtained from the SVM predictive model are listed in Table 1. Fig. 3 shows the predicted versus experimental values of pK_i for the training and test sets with SVM method. Table 3 presents the statistical parameters of the results obtained from the two studied models for the same set of compounds. The RMSE of SVM model for the training and test data sets was lower than those of models proposed in the MLR method. The correlation coefficient (R^2) given by SVM was higher than that of the MLR. The results of the F -test were obtained and also are shown in Table 3. From the table, it can be seen that SVM model

gives higher F values, so this model gives the most satisfactory results, compared with the results obtained from MLR method. Consequently, this SVM approach currently constitutes the most accurate method for prediction of the binding affinity of piperazinecyclohexanes.

4. Conclusion

In this study, MLR and SVM were used to develop linear and nonlinear QSAR models for prediction of melanocortin-4 receptor binding affinities of piperazinecyclohexanes. The built models clearly demonstrate good correlations between the structure and activity of the studied compounds. The validation of the models using the LOO and LGO cross-validation, external test set, and chance correlation, showed that the designed models have a good external predictive power. Comparison between MLR and SVM methods demonstrates that the performance of SVM model is better than that of MLR, which indicates that the binding nonlinear model is able to describe the relationship between the structural descriptors and the process more accurately. The proposed models can identify and provide an insight into some instructions for further synthesis of high-affinity melanocortin-4 receptors.

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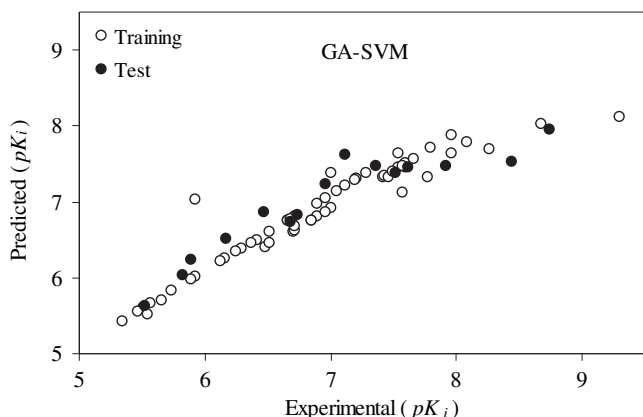


Fig. 3. Observed vs. calculated pK_i for the GA-SVM model with selected descriptors.

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